Unequal Job Opportunities and the Measurement of Welfare Inequality

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France 2016, EU-SILC priced through EUROMOD.

Two questions.

(1) How much money-metric well-being inequality is attributable to unequal job opportunities rather than to preferences?

(2) When a model omits heterogeneous opportunities, how much of that inequality is re-attributed to preferences?

Three literatures, one gap.

Money-metric welfare with heterogeneous preferences

Fleurbaey and Maniquet (2005, 2011); Decoster and Haan (2015); Bargain, Decoster, Dolls, Neumann, Peichl and Siegloch (2013); Jacquet, Jia and Thoresen.

Random-utility random-opportunity labour supply

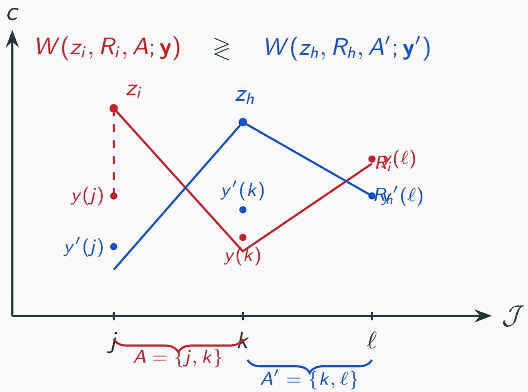
Dagsvik (1994); Aaberge, Colombino and Strøm (1999); Aaberge and Colombino (2013); Dagsvik and Jia (2016); Capéau et al.

Inequality of opportunity

Roemer (1998); Bourguignon, Ferreira and Menéndez (2007); Ferreira and Gignoux (2011).

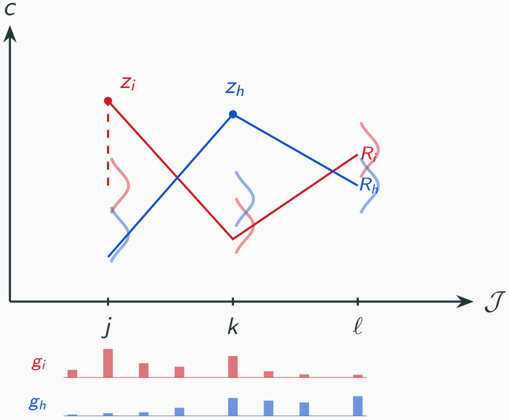
This paper: household-specific latent job-opportunity distributions carried into a preference-respecting money-metric inequality decomposition, in both counterfactual directions.

A job is a package; welfare is compared on the set of reachable jobs.



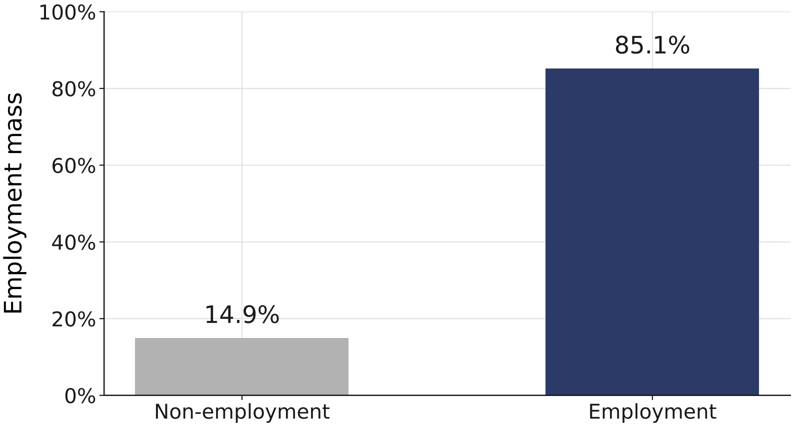
W[1]: the uniform pay across the jobs you can reach that would leave you as well off - Haydar and Maniquet (companion paper).

Empirically the set is an estimated distribution and the pay is an offer density.



estimated jointly with preferences; nodes are integration support, not a list of jobs.

For one household: its estimated employment mass, hours weights, occupation shares and wage-offer density



estimated opportunity distribution; nodes are integration support, not a list of jobs

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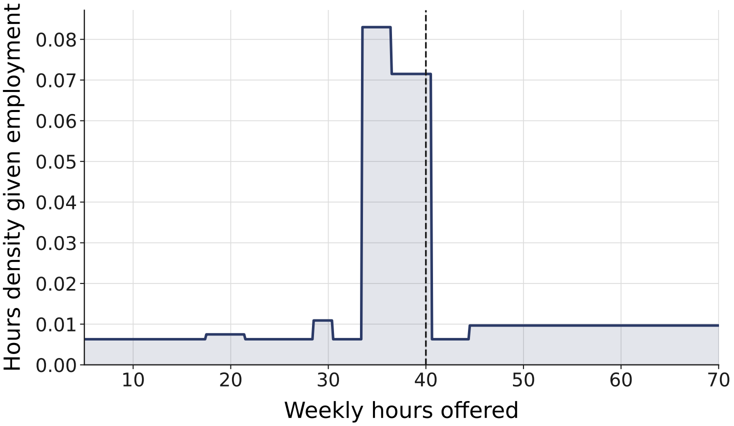

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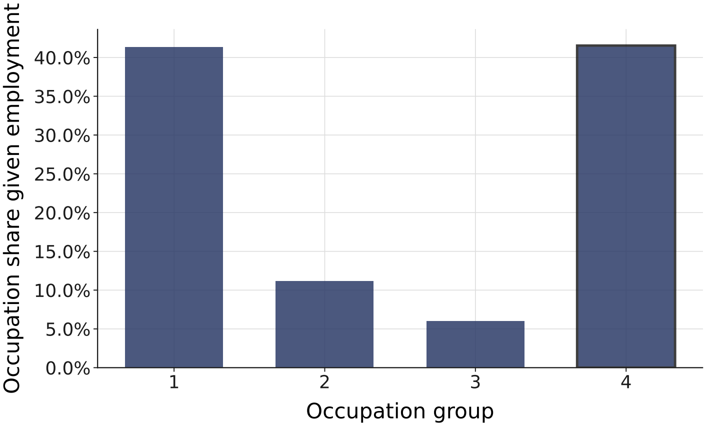

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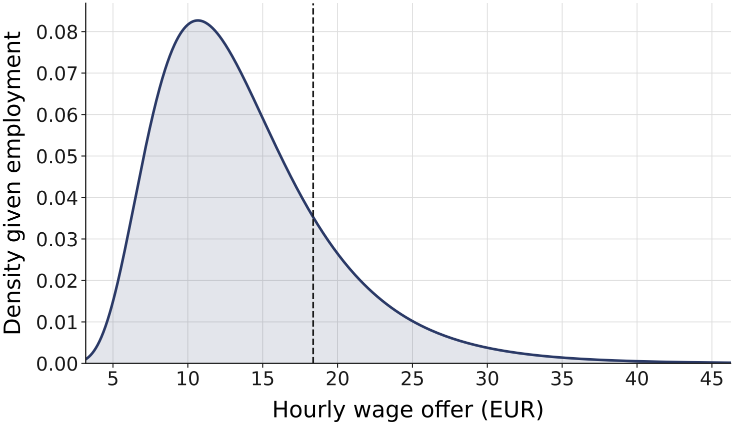

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Preferences and an opportunity density, in one likelihood.

$$u_{ij} = \beta_\ell(x_i)\,\mathrm{BC}(l_{ij};\theta_\ell) + \mathrm{BC}(c_{ij};\theta_c)$$

$$g_{ij} = \underbrace{g^E}_{\substack{\text{whether work}\\ \text{is available}}} \cdot \underbrace{g^H}_{\text{at which hours}} \cdot \underbrace{g^{\mathrm{Occ}}}_{\text{in which occupation}} \cdot \underbrace{g^W}_{\text{at what pay}}$$

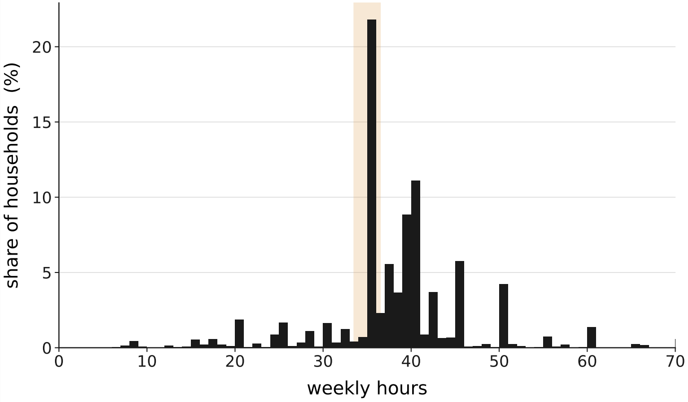
41 estimated structural parameters.

Each household: its observed job plus 100 drawn alternatives, all priced through the tax-benefit system.

$$V_{ij} = \underbrace{u_{ij}}_{\text{preferences}} + \underbrace{\log g_{ij}}_{\text{availability}} - \underbrace{\log q_{ij}}_{\text{sampling correction}}$$

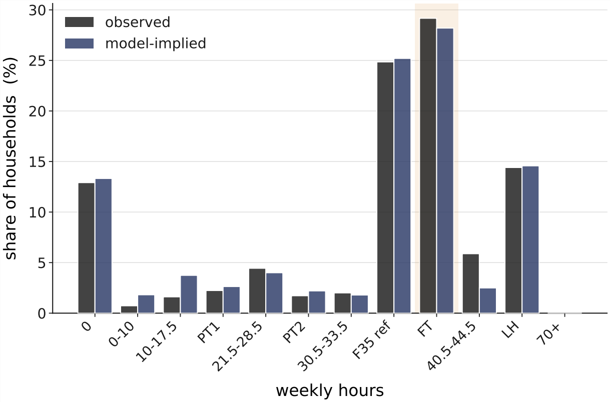
EUROMOD FR 2016 $\rightarrow$ disposable income for every one of 157,055 alternatives.

France 2016: 1,555 single-adult households; one in four employed at exactly 35 hours.



EU-SILC 2016; weights; 1,348 employed.

The model reproduces the hours distribution, employment and occupation shares.

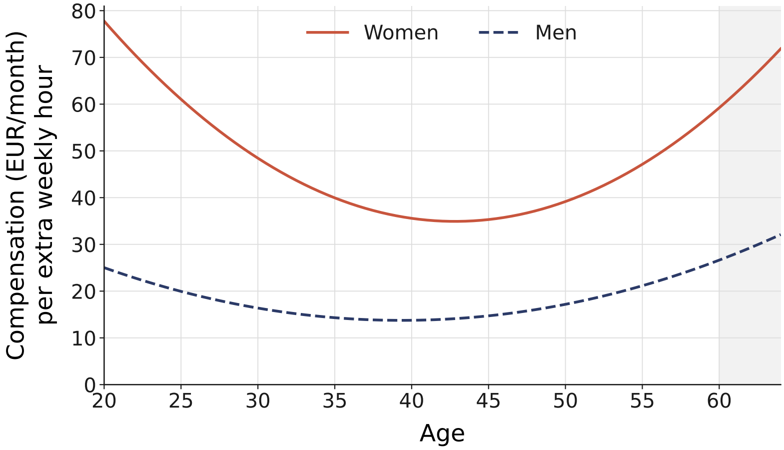


35-hour band 24.9% observed, 25.2% model-implied; employment 87.1% / 86.7%; mean absolute band error 0.008.

Estimated indifference curves and the price of an hour, by age and sex

Box-Cox preferences at the estimated parameters; own budget; one household per sex

Estimated indifference curves and the price of an hour, by age and sex



Box-Cox preferences at the estimated parameters; own budget; one household per sex

What the data pin down, and what remains conditional.

pinned

the preference block;

the four availability factors;

the 35-hour peak.

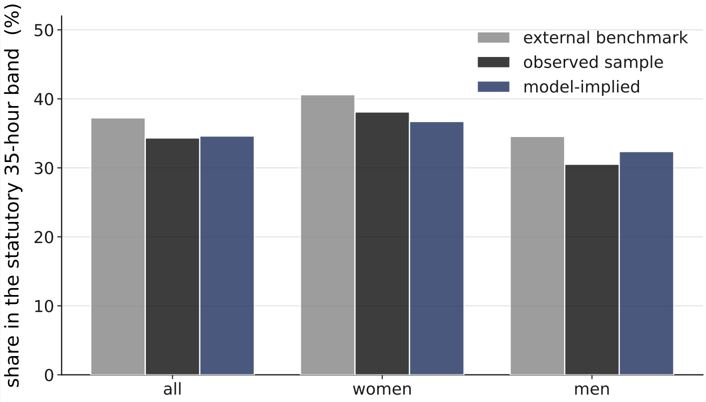
conditional

access versus capability (not separately identified);

persistent unobserved heterogeneity (three pre-specified extensions not recoverable from one cross-section).

structural and model-conditional; not causal.

An independent source puts the 35-hour concentration at 37%; the sample at 34%; the model at 35%.



Labour Force Survey 2016; validation only; no external moment enters the likelihood.

Equivalent income at a common reference pay, under four preference–environment states.

$$W_i^1 = w : \quad \Omega_i(\text{flat pay } w \text{ on own set}) = \Omega_i(\text{actual})$$

i.e. the uniform pay w over the household's own opportunity distribution that reproduces its attained expected welfare.

	own environment	reference environment
own preferences	I_{00}	I_{01}
reference preferences	I_{10}	I_{11}

reference environment = common job access, earning opportunities and budget; the same coalition's set and preferences on both sides of the inversion; common quadrature support.

A two-player Shapley game: preferences against the complete non-preference environment.

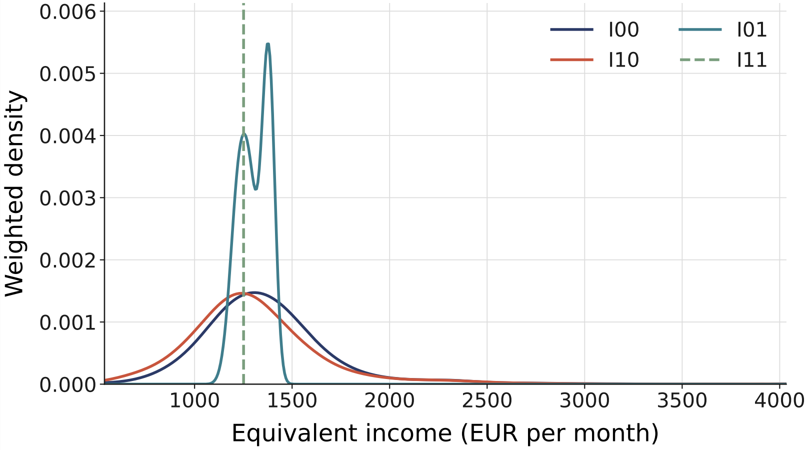
$$I_{00} = 0.134 \qquad I_{01} = 0.031$$
$$I_{10} = 0.148 \qquad I_{11} = 0.000$$

$$C_{\text{pref}} = \tfrac{1}{2}[(I_{00} - I_{10}) + (I_{01} - I_{11})] = 0.0085$$
$$C_{\text{env}} = \tfrac{1}{2}[(I_{00} - I_{01}) + (I_{10} - I_{11})] = 0.126$$

equalizing the environment alone reduces inequality by 77%, equalizing preferences alone raises it by 10%; the Shapley attribution averages the two orders and gives the environment 94%.

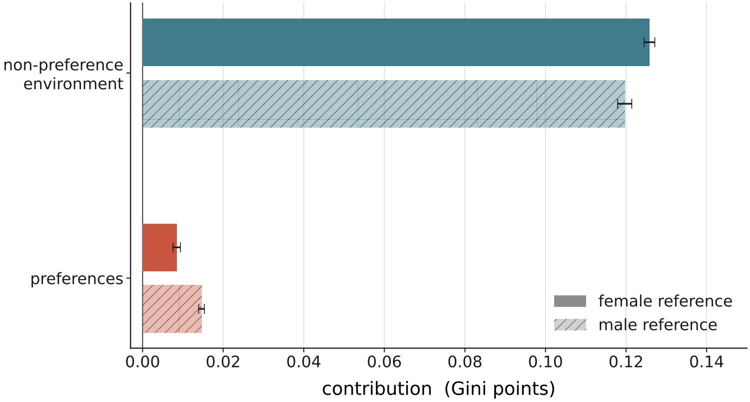
the environment is then split by an Owen value into job access, earning opportunities, and household endowments and needs.

The distribution of equivalent income under the four states



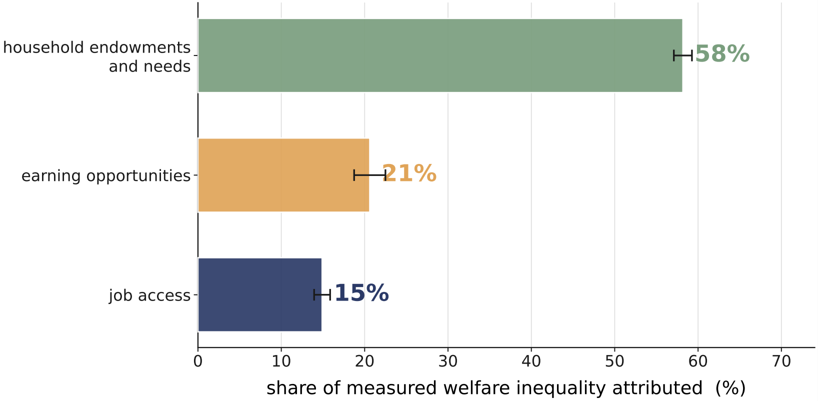
the fully common state collapses to a point

The non-preference environment accounts for 94% of measured welfare inequality; preferences for 6%.



93.7 $\pm$ 0.6 / 6.3 $\pm$ 0.6 female-primary; 89.1 / 10.9 male structural-zero; never averaged. Parameter intervals: environment [89.1, 95.8], preferences [4.2, 10.9].

Household endowments and needs 58%, earning opportunities 21%, job access 15%; job opportunities together 36%.



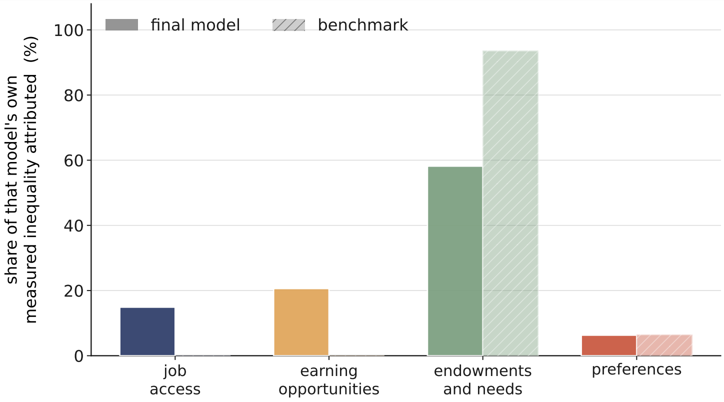
integration bands $\pm$ 1.5 / $\pm$ 1.6 / $\pm$ 1.1; ordering identical under both reference conventions and under equivalization.

Unequal job opportunities account for about a third of measured welfare inequality; omitting them re-attributes almost nothing to preferences.

(1) job access + earning opportunities: 35.5% $\pm$ 1.0 of measured inequality (the complete environment 93.7% $\pm$ 0.6).

(2) Omitting heterogeneous opportunities: preference share 6.3% $\rightarrow$ 6.4%; measured inequality falls 24.2%; the market-side contribution is relabelled into endowments and needs.

The benchmark fits the marginals as well, is 129 log-points worse, and recovers availability as taste.

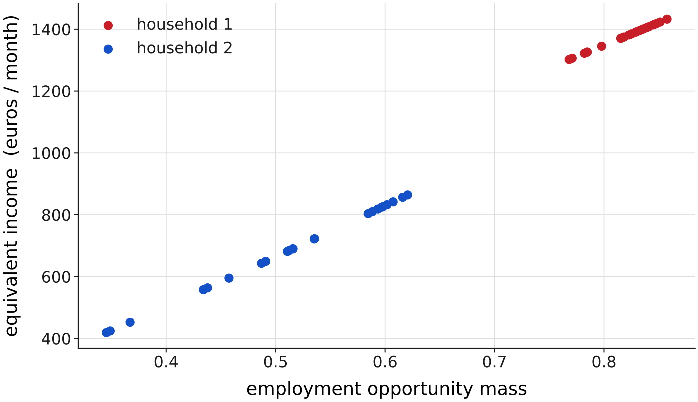


35-hour coefficient: 2.58 as availability, 2.55 as preference; leisure-intercept sex gap $+0.43 \rightarrow -1.99$.

The preference share is the sensitive margin; the environment's internal structure is stable.

perturbation $\rightarrow$	preference share	environment ordering
reference convention	$6.3 \rightarrow 10.9$	unchanged
age-curvature bounds	$\approx +13\%$	unchanged
couples male-leisure pin	up to 73%	unchanged
50-400 drawn jobs	largest coefficient move 0.56 s.e. (0.2 s.e. from 100 to 400)	ordering unchanged

Within job access, nearly all of the household-varying contribution is geographic; job access matters twice as much for men.



geographic access 13% of inequality, job access 15%; men 20%, women 10%; structural, not causal.

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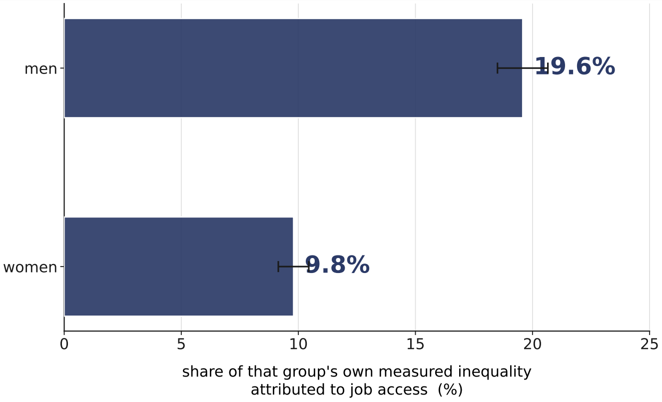

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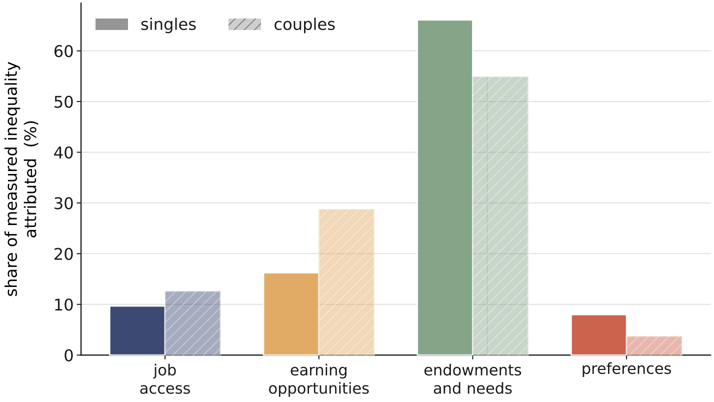
Couples reproduce the same environment ordering; their preference share is not robustly identified.



2,275 couples; employment matched to four decimals; 35-hour peak 2.55 (women) and 3.29 (men). Cross-leisure interaction absent; welfare runs it at exactly zero — spouse leisure complementarity is not estimated here.

Each limit names the evidence that would lift it.

access vs capability	$\rightarrow$	skill and credential data
persistent heterogeneity	$\rightarrow$	panel choices or external moments
hours opportunities	$\rightarrow$	desired-hours moments with a model counterpart
earning opportunities	$\rightarrow$	administrative wage distributions
geography	$\rightarrow$	local vacancy data

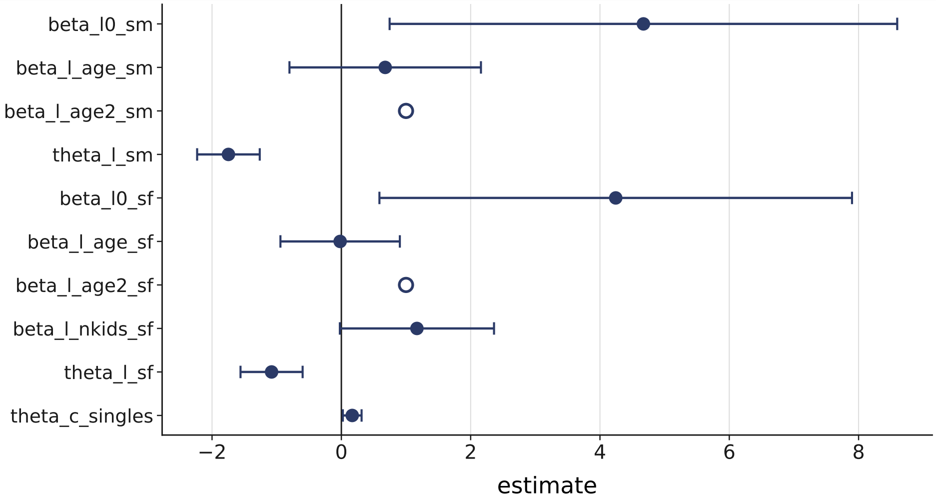
Conclusion

Method: a household-specific opportunity distribution estimated with preferences, priced through the real tax-benefit system.

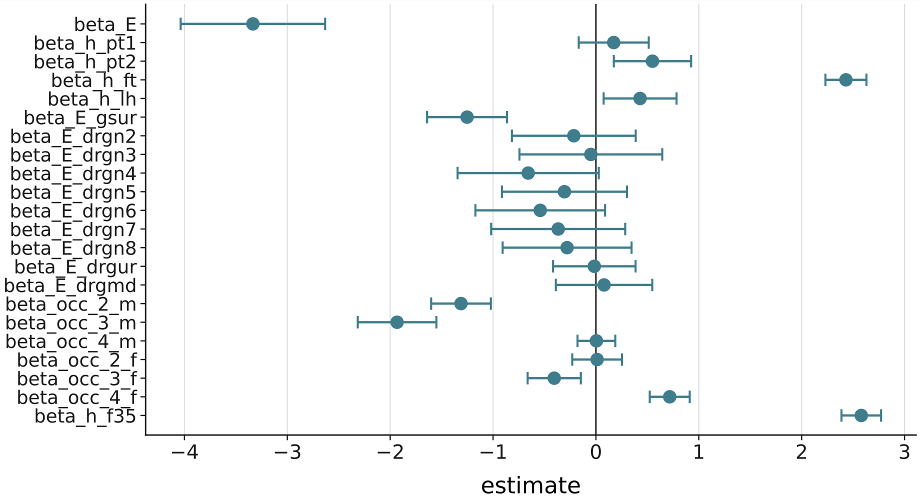
Finding: the non-preference environment carries 94% of measured money-metric inequality; job opportunities about a third; the budget side more than the market side.

Margin: the preference share is the sensitive quantity, 6 to 11 per cent across normative conventions.

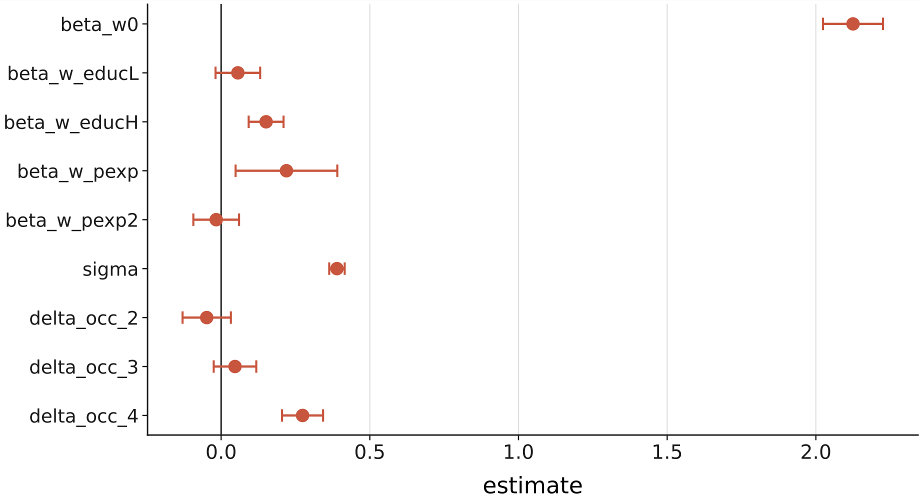
B1 — The estimated coefficients: preferences



B1 — The estimated coefficients: job access



B1 — The estimated coefficients: earning opportunities



B2 — Exhaustiveness is a tested property

preferences	job access
earning opportunities	household endowments and needs

The fully common state is 0.000000 — numerically zero.

B3 — The scale moves with the channel that equalises needs

	raw	equivalized
preferences	6.30%	7.98%
opportunities and budgets	93.70%	92.02%
job access	14.90%	9.68%
earning opportunities	20.62%	16.24%
household endowments and needs	58.18%	66.10%
market side	35.51%	25.92%

Freezing the own scale leaves exactly the Gini of its inverse.

B4 — the desired-hours counterpart test

ground	what it shows
definitional	the candidate's optimum tracks pay
identification	the hours block moves the moment by **exactly 0**
numerical	the level can agree; the profile cannot

Tried, and it fails on three independent grounds.

B5 — Where the welfare family comes from

Compensation

unequal productivity should not become unequal well-being

Responsibility

preference-based choices should not be compensated

$$\nexists\, W : \quad \text{Full Compensation} \;\wedge\; \text{Full Responsibility}$$

[Fleurbaey & Maniquet, *SCW* 2005]

W^1 is one compromise — the one this paper takes to the data.

Haydar and Maniquet, *Jobs and Well-Being Measurement* (companion paper).

B6 — The RUM benchmark details

	final model	benchmark
35-hour coefficient	2.58	2.55
	as availability	as preference
leisure-intercept sex gap	+0.43	-1.99
preference share	6.3%	6.4%

Job access and earnings are exactly zero here. By construction.